

Prefill Once, Ask Many: Training-Free Reversible Below-Page KV Selection for Multi-Query Long-Document Visual Question Answering

Anonymous submission

Abstract

Real document-VQA workloads are multi-query: one long document is prefilled once and asked many different questions, but standard KV-cache eviction commits to one compressed context for all questions. We present PAQ-KV (“prefill once, ask many”), a training-free architecture for per-query re-selection over a retained full-context KV cache: prefill once, score 64-token page-aligned blocks by the reader’s own question-to-content attention, and decode with a restricted-attention 4D mask over the intact cache (no gather, no re-read, no second model, no index). Because nothing is discarded, selection is fully reversible—the property the architecture unlocks—and per-query re-selection beats the identical scorer committed once, a strong gold-informed static, and a trained retriever’s committed form (isolation contrast $+0.264$, $LB_{95} = +0.187$ under a clean-cache re-run; block-level replication $+0.379$, $LB_{95} = +0.273$). At a nominal 5% (realized $\approx 6\%$) decode-attention budget, PAQ-KV beats ColQwen2, a state-of-the-art trained visual retriever, on MMLongBench-Doc (0.4042 vs. 0.3282 with ColQwen2’s trained retrieval-projection LoRA correctly loaded; $+0.077$, document-clustered $LB_{95} = +0.040$; 80 documents, 444 decoded / 440 paired), matches full context, and is *not statistically distinguishable* from the gold-page oracle at its $\sim 6\%$ budget (-0.026 , CI straddles zero—parity within noise, not a demonstrated equality). On LongDocURL PAQ-KV matches but does not beat the retriever ($+0.019$, $LB_{95} = -0.019$; below full context by -0.014); we report this as a limitation and observe the cell appears largely reader-conversion-bound rather than selection-bound. Two selector upgrades (a per-head mixture transfer gate that stopped the LoRA lane before any adapter was trained; a sharp expert-head scorer) failed pre-registered gates and are reported as negatives. The beat is **confirmed out-of-sample**: on a fresh, disjoint, sealed 45-document / 244-query MMLongBench-Doc set evaluated exactly once under a pre-registered frozen harness, PAQ-KV@5% scores 0.396 vs. ColQwen2 0.320 ($+0.076$, document-clustered $LB_{95} = +0.029$; 244/244 complete, robust to worst-case imputation).

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1 Introduction

Long-context vision-language models (VLMs) can read an entire multi-page document, but they pay for every page on every question. The standard efficiency response, inherited from single-query text benchmarks, is to *evict*: keep a compact subset of the KV cache and decode against it (Li et al. 2024; Zhang et al. 2023; Chen et al. 2024; Wan et al. 2024; Kim et al. 2025). Eviction is a single, irreversible commitment: whatever subset is kept must serve every future question about that document.

Document assistants, however, are not asked one question. The two long-document VQA benchmarks we use are natively multi-query: grouping their 64K splits by source document, MMLongBench-Doc (Ma et al. 2024) averages ≈ 6.97 questions per document and LongDocURL (Deng et al. 2025) ≈ 3.6 ; inside our pinned evaluation subsets the averages are 5.50 and 5.79 gold-bearing questions per document. Different questions need different evidence, so a subset committed for question 1 is *stale* for question 5.

This paper asks: at a matched budget on long multimodal documents, does per-query *re-selection* of context beat a *commit-once* selection—and can a training-free, reader-internal selector compete with a dedicated trained retriever? Our answer is PAQ-KV: prefill the full document *once* into the KV cache of a frozen reader (Qwen3-VL-8B; Qwen Team 2025) and retain it; per query, re-select the most relevant 64-token, below-page KV blocks with the reader’s own question→content attention; decode under a restricted-attention 4D mask over the retained full-context cache. Nothing is discarded and nothing is gathered (M-RoPE positions are untouched), so a selection is a *view*, not a commitment: the next question re-selects from the same intact cache.

Contributions.

- **A reversible below-page architecture and its tested mechanism.** PAQ-KV keeps the full-document prefill intact and, per query, re-selects 64-token below-page KV blocks and decodes under a 4D restricted-attention mask *without gathering* the cache, so all M-RoPE positions stay the model’s own. Under a pre-registered two-comparator gate, per-query re-selection beats the identical scorer committed once ($+0.257/+0.163$ per cell; 903 queries, 80 documents/cell, document-clustered) and a strong gold-informed static ($+0.108/+0.104$); the con-

trast survives a clean-cache re-run (+0.264, $LB_{95} = +0.187$), reappears below page level (+0.379, $LB_{95} = +0.273$), and also holds for a trained retriever’s committed form.

- **A sealed-confirmed beat, with explicit bounds and negatives.** On MMLongBench-Doc, PAQ-KV at a nominal 5% (realized $\approx 6\%$) budget scores 0.4042 vs. ColQwen2 0.3282 (+0.077, $LB_{95} = +0.040$; 80 documents, 444 decoded / 440 paired; ColQwen2 evaluated with its trained retrieval-projection LoRA correctly loaded, §4.4), is indistinguishable from full context, and is not statistically distinguishable from the gold-page oracle at its $\sim 6\%$ budget (-0.026 , CI straddles zero); the beat also holds on a fresh, disjoint, sealed set (+0.076, $LB_{95} = +0.029$). On LongDocURL PAQ-KV matches but does not beat ColQwen2 (+0.019, $LB_{95} = -0.019$; below full context by -0.014), and both the per-head mixture transfer gate and the sharp expert-head scorer failed pre-registered gates.

Evidence status. Primary full-method accuracy results use the 80-document-per-cell development pins unless explicitly labelled otherwise; the mechanism, budget, and integrity analyses use the stated subsets (30/24/50-document and 157-query screens). Only the MMLongBench-Doc PAQ-KV-versus-ColQwen2 comparison was evaluated on the sealed set—once, on a fresh, disjoint 45-document / 244-query cell, where the beat holds (+0.076, $LB_{95} = +0.029$; §5.2). The sealed LongDocURL documents remain unevaluated (out of scope for this MMLongBench-scoped result). We state the evidence status wherever it binds.

2 Related Work

Page selection and retrieval. CAPS (Zhu et al. 2026) publishes attention-based *page* selection on our benchmarks, and ColPali/ColQwen2 (Faysse et al. 2025) are the strongest trained retrieval baselines we compare against. We state plainly that “reader-attention selects pages” is CAPS’s method; PAQ-KV differs on the axes CAPS names as open: below-page granularity, reversible per-query re-selection over one retained prefill in the multi-query regime, and direct masked decode from the retained cache with no re-read.

Eviction context and novelty map. SnapKV (Li et al. 2024), H2O (Zhang et al. 2023), FastV (Chen et al. 2024), LOOK-M (Wan et al. 2024), and KVzip (Kim et al. 2025) commit a compressed cache once; Quest (Tang et al. 2024) shows query-aware selection can beat query-agnostic eviction in single-query text, motivating but not proving our multi-query claims. Our search did not identify prior work in the exact combined cell—reader-internal attention as the runtime below-page selector, end-to-end QA at matched budget against page-level retrieval, on long multi-query documents—which we offer as a provisional literature-map claim, not proof of novelty.

3 Method: PAQ-KV

3.1 Setting

One long visual document, consumed as page images (no OCR), is read by a frozen VLM and will be asked k questions (5.5–5.8 gold-bearing per document in our pins). The document is prefilled **once** at the same rendering the full-context arm uses (a nominal $\approx 64K$ -token target; realized prefill lengths reach $\approx 92K$ on the largest pinned documents) and the KV cache is *retained*: never evicted, never physically compressed, never gathered. The cache plays the role a multi-vector index plays for a retriever—except it is the reader’s own working memory, so retrieval and reading share one representation and one model.

3.2 Per Query: Preview, Re-Select, Masked Decode

1. **Preview (score).** The tokenized question is forwarded over the retained cache—with the cache crop-restored afterwards, so queries never contaminate one another (§4.4)—and the question rows’ attention over context positions is aggregated into scores over **64-token, page-aligned KV blocks**: fine enough to isolate a table row, a figure caption, or a paragraph; coarse enough for the scores to be stable.
2. **Re-select.** Keep the top-scoring blocks at a **nominal 5% decode-attention budget**. The terminology is deliberate: the kept blocks’ K/V were computed with full-document context during the prefill (the standard claim semantics of the Quest/SnapKV method class), so this is a decode-attention budget, not a matched-token prefill comparison; the realized attended fraction on the executed screen is $\approx 5.9\text{--}6.0\%$ of context and is always reported. The matched quantity against ColQwen2 is the evidence budget (5% of pages vs. 5% of context) at the same shared reader and QA evaluator; the two methods’ *selection* scorers differ (reader attention vs. the trained ColQwen2 retriever).
3. **Masked decode.** The answer is generated while attending only to the selected blocks plus a small **always-keep set** (the BOS/system/template scaffold, the question itself, and previously generated tokens), enforced by a **4D additive attention mask** (batch, head, query row, key) over the retained cache. **No gathering**: the K/V tensors stay at their original positions, so every M-RoPE rotary index is exactly the model’s own—the failure mode that makes physical cache-gathering treacherous for VLMs is sidestepped entirely. No re-read, no re-render, no second model.

Algorithm 1 (Appendix E) gives the loop and Figure 1 shows the mask machinery. Attention is captured by OOM-safe per-layer forward hooks that reduce each layer’s attention tensor immediately and discard it, so peak memory is one layer’s attention (the naive all-layers path OOMs at long contexts).

3.3 Exactness and Reversibility

A decode-identity audit checked the mask path against the native decode over 6 document–query pairs deliberately

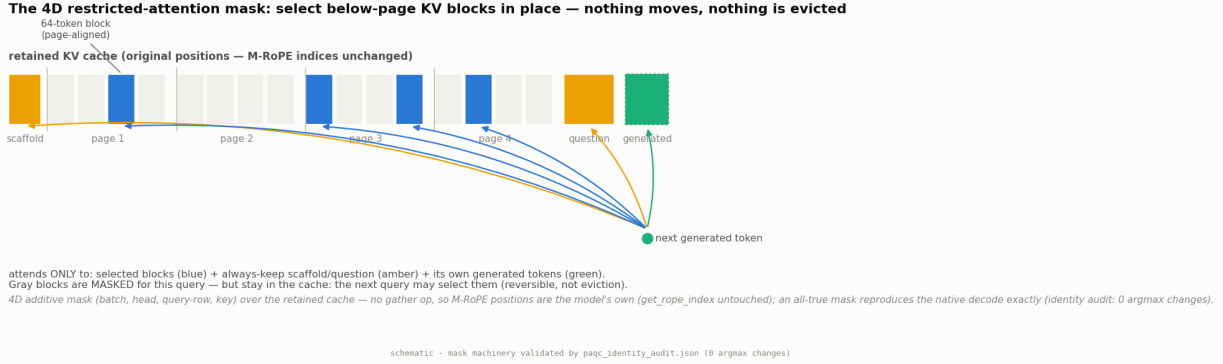


Figure 1: The mechanism. The retained cache is a strip of K/V at their original positions: scaffold, page content in 64-token page-aligned blocks, question, generated tokens. The 4D additive mask lets decode rows attend only to the per-query selected blocks plus the always-keep set; everything else is masked *for this query only*—it stays in the cache and remains selectable by the next query. Because nothing is gathered, the M-RoPE position machinery is untouched, and an all-true mask reproduced the native decode in every audited case (identity audit: zero mask-specific argmax changes over the audited pairs).

spanning 32K–122K tokens (a stress range wider than the $\leq 64K$ evaluation pin, to probe long-context behavior): in **all 6 audited cases** an all-true mask produced **zero mask-specific argmax changes** relative to the native causal decode (an earlier single greedy-token divergence traced to the generation loop’s mask-independent fast-path kernel, not to the mask). We therefore claim the mask decode is identical to the native decode in the audited cases, not a general proof; this audit was a pre-registered kill criterion. Because nothing is discarded, a selection is a view, not a commitment: the next query re-scores and re-masks the same complete cache, whereas commit-once eviction cannot—what it dropped is gone. Reversibility is our tested axis (§4.2), and the contrast replicates at block level ($+0.379$, $LB_{95} = +0.273$; §5.2).

What PAQ-KV is not. It is not CAPS (a separate scoring VLM, per-page full re-forwards, page granularity, single-query); not coarse-to-fine page routing with a full-resolution re-read (our own earlier page-level system, Appendix B); and not eviction (the mask is per-query and reversible). The mask route is the *accuracy* path; physically gathering the cache for savings is a separate, future efficiency claim we do not make.

4 Experimental Setup

4.1 Benchmarks, Pins, Reader

We evaluate on **LongDocURL** (Deng et al. 2025) and **MMLongBench-Doc** (Ma et al. 2024), both as page images. For each document we build the union of the benchmark’s own-document pages across that document’s queries (dropping cross-document 64K padding), so every query’s evidence is present in one prefill. Documents qualify with ≥ 3 gold-bearing queries and ≤ 110 union pages (a nominal $\approx 64K$ -token target at read resolution; realized lengths reach $\approx 92K$ on the largest documents); selection is seed-0 frozen. The confirmatory pin holds 80 documents per cell with 463/440 complete queries and 5.79/5.50 gold-bearing queries per document (LongDocURL/MMLongBench-Doc; Appendix E); a 30-document screen pin is its strict prefix,

and results are additionally checked on the 50 new documents only (§5.1). The reader is **Qwen3-VL-8B** (Qwen Team 2025), frozen (SDPA attention for prefill and reading; eager attention only during the per-query attention preview, so question→content scores can be captured). Page-level arms operate at a matched page budget $b \in \{2.5\%, 5\%, 10\%\}$ (the pre-registered gate budget is $b = 5\%$); accuracy is the benchmark’s document-VQA scorer. Everything in this paper is a Qwen3-VL-8B case study; no second *reader* VLM has been evaluated.

4.2 Statistical Protocol

All inference uses a **document-cluster paired bootstrap**: queries of one document share the document and (for stale arms) the single committed subset, so bootstrapping over queries would be anti-conservative pseudoreplication. Intervals resample documents with replacement within each cell (10,000 draws, seed 0); LB_{95} denotes the 2.5th percentile of the bootstrap distribution—the lower end of a two-sided 95% CI. Per-cell results are primary; pooled contrasts use equal cell weighting and are labelled. Every gate in this program was frozen before its run.

The isolation gate. “Query-aware beats query-agnostic” is both a known text result (Tang et al. 2024) and a strawman risk. We therefore fix the tested axis to **reversibility**: every baseline runs its canonical scorer but is deployed *stale*—committed once per document (query-aware scorers commit a min-max-normalized aggregate over the document’s first $M = \min(3, k)$ queries)—while the fresh arm re-selects per query. Reversibility is credited only if the fresh arm beats **both** (i) its *own* scorer committed once (identical scorer and budget, so the delta isolates the mechanism) and (ii) a strong *gold-informed* static subset (top pages by gold-page frequency across the document’s queries—not deployable, and not provably optimal, but immune to the weak-static artifact that invalidated an earlier one-comparator gate).

4.3 Baselines

The executed panel (Appendix A) compares, per query at matched budget on the same pinned data and reader: **ColQwen2** (Faysse et al. 2025) fresh and stale, **CLIP** (Radford et al. 2021) fresh, SnapKV fresh/stale, the isolation arms above, random/truncation controls, and a demoted query-blind eviction family (H2O; a context-only FastV proxy; LOOK-M; a KVzip `recv_max` proxy)—disclosed proxies that sit at or below random page recall here and are never claimed as beaten opponents. Full-context and gold-page-oracle rows are references, not competitors.

4.4 Methods Integrity

A cross-family build review found a real bug in the per-query scoring loop used across this program: the runtime mutates the retained cache in place, and the loop reused the document prefill across a document’s queries without restoring it, so query q_i scored while attending to $q_0 \dots q_{i-1}$. The fix (crop the cache after every per-query preview) is now applied everywhere, and its impact was quantified rather than assumed: on a 24-document/150-query clean-vs-polluted re-run, the key isolation contrast is $+0.264$ ($LB_{95} = +0.187$) clean vs. $+0.284$ polluted—the contrast largely cancels the bug because both arms share the same scorer—while absolute fresh-arm accuracy carries ≈ 0.04 inflation in the (page-level, pre-fix) confirmatory run, which we disclose wherever absolute levels matter. All PAQ-KV results postdate the fix.

Baseline-loader integrity. A second cross-family review caught a defect in the *comparator*, not our method: the cached ColQwen2 baseline had run with its trained final retrieval-projection LoRA (`custom_text_proj`) silently zeroed. `vidore/colqwen2-v1.0` is a PEFT adapter, and under our `peft/transformers` versions a key-remap inserted `language_model` into the LoRA target path—correct for the language-model layers but wrong for the top-level `custom_text_proj`—so the checkpoint’s projection-LoRA tensors landed at a non-matching path and the final projection ran with a zero LoRA delta (base weights only). Because the other adapter layers still loaded, the baseline was functional (well above random), which masked the defect until we inspected the load report and the projection weights directly. We patched the loader (copy the two checkpoint `custom_text_proj` LoRA tensors onto the correct module; a post-load assertion now verifies they match the checkpoint) and **re-scored ColQwen2 with the corrected loader**. The correction is *immaterial* to the headline: fixed-vs-broken gold-page recall@5% is 0.539 vs. 0.534 ($+0.005$), the top-5% page selections overlap at Jaccard 0.947 (only 34/444 queries change any selected page), and re-reading those changed queries moves fresh-read accuracy by -0.0002 (0.3282 corrected vs. 0.3284 broken); the beat is $+0.077$ ($LB_{95} = +0.040$) against the corrected ColQwen2 and $+0.077$ ($LB_{95} = +0.039$) against the broken one. The trained projection LoRA does not materially reorder ColQwen2’s rankings on this MMLongBench-Doc data, so the headline is not loader-inflated. The corrected loader is used for the **MMLongBench-Doc headline (DEV and sealed) and the systems measurement**; the Long-

DocURL hold and the historical page-level panel retain the earlier (uncorrected) loader and are *labelled as such*. We do not infer a direction for the correction on the cells that were not re-run: on MMLongBench-Doc the change was negligible in both directions (gold-recall $+0.005$; fresh-read QA -0.0002), and LongDocURL—already a matched hold, not a beat—is reported as an uncorrected-loader development result.

Separately, the program adopted a **sealed-test discipline**: the 80-document pin is development-only, and a fresh, disjoint 45-document / 244-query MMLongBench-Doc set (zero overlap with the pin) was reserved and evaluated *exactly once* under a frozen harness—the confirmatory bar, comparator, budget, imputation policy, and all eight component code hashes were frozen (and a byte-hash of the sealed pin recorded) before the single run, which a guard enforces. That one sealed evaluation **confirms the beat out-of-sample** (§5.2); the sealed LongDocURL documents were not evaluated (out of scope for this MMLongBench-scoped result).

5 Results

5.1 Reversibility Is Real

At $b = 5\%$ on 903 complete queries (80 documents/cell), the fresh page-level selector scores 0.386/0.258 (LongDocURL/MMLongBench-Doc) vs. 0.128/0.095 for the *identical scorer committed once* and 0.278/0.154 for the gold-informed static: fresh – commit-once is $+0.257/+0.163$ and fresh – static is $+0.108/+0.104$ (both cells clear; full table in Appendix E). Figure 3 shows every executed isolation interval. Figure 3 (appendix) shows the tested axis schematically together with every executed isolation interval. Three robustness checks. (i) *Independent extension*: because the screen pin is embedded in the confirmatory pin, we re-ran the statistics on the 50 new documents per cell only ($n = 582$ queries): fresh – commit-once $+0.216$ [$+0.159, +0.269$]; fresh – static $+0.108$ [$+0.059, +0.150$]. (ii) *Clean-cache re-run*: the contrast survives the integrity fix of §4.4 ($+0.264$, $LB_{95} = +0.187$). (iii) *Scorer generality*: the same fresh-over-stale collapse holds for a retrieval scorer—ColQwen2 fresh 0.423 vs. stale 0.178 (query-micro; equal-cell 0.420 vs. 0.177)—so reversibility is a property of the multi-query regime, not of our scorer. A pre-registered “commitment-pressure” mechanism gradient *failed* its formal interaction test (both intervals include zero) and is retained as exploratory only; we make no mechanism-gradient claim.

5.2 MMLongBench-Doc: The Beat, and Its Audit

The result. On the 80-document dev pin (444 queries decoded; contrasts on the 440 with cached references), PAQ-KV@5% scores 0.4042 vs. ColQwen2-fresh 0.3282: $+0.077$, $LB_{95} = +0.040$ —the pre-frozen per-cell bar clears with margin (Table 1). ColQwen2 here is loaded with its trained retrieval-projection LoRA correctly applied; §4.4 documents a loader defect we found and corrected, and shows the correction is immaterial to this beat.

Out-of-sample confirmation (the sealed one-touch). The beat holds on a *fresh, disjoint, sealed* set evaluated

	LongDocURL	MMLongBench-Doc
PAQ-KV@5% (ours)	0.5315	0.4042
ColQwen2-fresh (matched budget)	0.5124	0.3282
full-context (reference)	0.5456	0.3965
gold-page oracle (reference; not strict)	0.6092	0.4314
PAQ-KV – ColQwen2-fresh	+0.019 [−0.019, +0.059]	+0.077 [+0.040, +0.115]
PAQ-KV – full-context	−0.014 (n.s.)	+0.009 [−0.022, +0.040]
PAQ-KV – gold-page oracle	—	−0.026 [−0.059, +0.008]
Queries (decoded / paired)	463 / 463	444 / 440
Verdict (pre-frozen bar: per-cell $LB_{95} > 0$)	HOLD	BEAT HOLDS

Table 1: PAQ-KV at the 80-document development pin, nominal 5% budget (all values on the same 80-document dev cohort; document-cluster paired bootstrap; contrast rows $\Delta [LB_{95}, UB_{95}]$). All four accuracy rows are the verified 80-document dev-pin means—no 30-document screen values are mixed in. On LongDocURL PAQ-KV is *below* full-context (−0.014, not significant); on MMLongBench-Doc it is indistinguishable from full-context. MMLongBench-Doc decodes 444 queries; the paired contrasts against the cached ColQwen2-fresh, full-context, and gold-oracle rows are computed on the 440 queries with cached references (the 4 mi_phone rows lack cached references).

exactly once. On 45 MMLongBench-Doc documents / 244 queries with zero overlap with the development pin—under a frozen harness whose bar, comparator, budget, imputation policy, and code hashes were all pre-registered (§4.4)—PAQ-KV@5% scores 0.396 vs. ColQwen2-fresh 0.320: +0.076, document-clustered $LB_{95} = +0.029$, complete (244/244, zero skips) and unchanged under worst-case missingness imputation. The pre-registered verdict (complete-case $LB_{95} > 0$, with an adversarial-imputation fallback that binds only if any query is missing) is met. The sealed effect (+0.076) closely matches the development-pin estimate (+0.077); the sealed interval is appropriately wider ($LB_{95} = +0.029$ vs. +0.040) at the smaller held-out sample. **Only the PAQ-KV-vs-ColQwen2 comparison was sealed-evaluated**—the full-context and gold-page-oracle reference arms above are development-pin only and were not run on the sealed set. This single out-of-sample evaluation is the program’s one mandate-grade claim, and it directly answers the reused-development-pin concern of §6. PAQ-KV attends to a nominal 5% (realized ≈ 5.9 –6.0%) of the context; at that budget it is indistinguishable from full context (+0.009 [−0.022, +0.040]) and **not statistically distinguishable from the gold-page oracle** (−0.026 [−0.059, +0.008])—the interval permits material inferiority, so this is parity within noise, not a demonstrated equality.

The mechanism audit. On the 30-document screen (distinct from the 80-document headline cohort), the beat survives every mechanistic adversarial check: query-dependent (PAQ-KV – random-block selection = +0.329 [+0.238, +0.418]), query-specific (PAQ-KV – wrong-question selection = +0.698 [+0.603, +0.784], where the collapse supports query-specificity), reversible (PAQ-KV – commit-once = +0.379 [+0.273, +0.480]), and below-page > page-level (+0.054 [+0.011, +0.102]). These are development-pin screen analyses with disclosed caveats: the screen budget-curve monotonicity gate failed on this cell (0.398 at 10% vs. 0.402 at 5%), and the beat they dissect is separately confirmed out-of-sample on the sealed set above.

Additional matched-budget baselines on the beat cell.

On the same 80-document pin and reader, a BM25 page retriever scores 0.2109 at matched 5% budget and PAQ-KV beats it by +0.193 ($LB_{95} = +0.152$; 444 queries), while a training-free *page-level* CAPS-style expert-head control scores 0.3561 and below-page PAQ-KV beats it by +0.048 ($LB_{95} = +0.009$; 444 queries; worst-case missingness imputation unchanged): the margin over this control is real but thin, so the beat over the *trained retriever* remains the stronger claim, and the screen-only budget-sweep details are moved to Appendix F.

5.3 LongDocURL: Matched, Not Beaten

On the 80-document dev pin (463 queries), PAQ-KV@5% scores 0.5315 vs. ColQwen2-fresh 0.5124: +0.019, $LB_{95} = -0.019$ —a positive point estimate that does **not** clear the pre-frozen bar; on this cell PAQ-KV also sits below full context (−0.014, not significant). The mechanism checks (query dependence, reversibility) clear here too; what is missing is the margin. We treat this as a bounding observation on where below-page selection helps, not a second headline, and move the conversion-bound decomposition to Appendix G.

5.4 Negative Ablations, at Full Prominence

Head-mixture transfer gate (the pre-LoRA gate that stopped the B lane): STOP. No LoRA adapter was trained: the first pre-frozen check fit a logistic **per-head mixture** over all $36 \times 32 = 1,152$ (layer, head) question→page attention channels on off-benchmark provenance-checked data. The transfer gate fails: mixture gold-page recall is 0.652/0.594 vs. 0.676/0.616 for the single head, with pooled mixture – single = −0.023 [−0.037, −0.008] (907 queries), so the mixture overfits and the lane was closed; the best zero-benchmark-contact head (**L21.H18**) still coincides with the benchmark-supervised expert layer.

Sharp expert head as block scorer: NO-GO. Replacing PAQ-KV’s diffuse mean-attention block scorer (V0) with the sharp head L21.H18 (V1) fails the pre-frozen gate: on LongDocURL it only nudges the point estimate (0.5315 → 0.536;

	extra params	per-doc state	online lat./q	peak VRAM
PAQ-KV (ours)	0	8.15 GiB	1.04 s	44.5 GiB
ColQwen2-fresh	2.246 B	11 MiB	0.46 s	21.8 GiB
full-context	0	—	11.59 s	44.6 GiB

Table 2: Deployment profile on an 8-document / 31-query systems subset (frozen configuration, single GPU). “Extra params” = additional retriever/selector parameters beyond the shared reader. PAQ-KV is $\approx 2.25\times$ slower and far heavier in resident state than ColQwen2, and $\approx 11\times$ faster than full context (cold-start amortized after $N \approx 1.07$ queries/document). PAQ-KV’s online latency excludes ≈ 1.99 s/query of removable research-harness input reconstruction (component sum ≈ 3.03 s/query in the current implementation); mean generated lengths differ across arms (5.3/3.5/4.1 tokens for PAQ-KV/ColQwen2/full-context). The win is 0 extra retriever parameters, no index, one model, and reversible query-time selection—*not* speed.

V1 – ColQwen2 = $+0.024 [-0.012, +0.066]$ and does not separate from V0 ($LB_{95}(V1 - V0) = -0.011$, n.s.), while on MMLongBench-Doc it hurts ($0.402 \rightarrow 0.381$ on the screen documents). As elsewhere in the program, **recall-optimized selection is not accuracy-optimized selection**.

5.5 Systems Accounting: Simplicity, Not Speed

We measured the deployment profile of the true PAQ-KV path—below-page masked decode over the *retained* full-document KV cache—against ColQwen2-fresh and full context on an 8-document / 31-query systems subset of the pin (single RTX PRO 6000; frozen configuration; Table 2); this corrects an earlier internal accounting that timed a different page-routing variant. Per online query, PAQ-KV costs 1.04 s (0.19 s selection + 0.85 s masked decode), *excluding* ≈ 1.99 s/query of removable research-harness input reconstruction (component sum ≈ 3.03 s/query), versus ColQwen2’s 0.46 s: PAQ-KV is $\approx 2.25\times$ slower and far heavier in resident state (≈ 8.15 GiB/document retained cache vs. ≈ 11 MiB index, $\approx 730\times$; peak VRAM 44.5 vs 21.8 GiB), though it is $\approx 11\times$ faster than full-context re-reading per query (1.04 vs 11.6 s; amortized crossover $N \approx 1.07$ queries/document). The efficiency claim is therefore *simplicity and deployment shape*, not speed: no separately trained retriever checkpoint (0 extra trained parameters vs. 2.246 B), no stored vector index, one deployed model, and reversible query-time selection.

6 Limitations

- **Reused development pin (adaptive-reuse caveat), with the headline confirmed out-of-sample.** Most PAQ-KV numbers come from the 80-document development pin, which has been *reused* across the program’s screening, auditing, and gating; pre-registered gates limit but do not eliminate the resulting adaptive-analysis/multiple-comparisons risk, so the development-pin *analyses and ablations* remain development-grade. The **headline**

MMLongBench-Doc beat is exempt: it is confirmed out-of-sample on the fresh, disjoint, sealed 45-document / 244-query set, evaluated exactly once ($+0.076$, $LB_{95} = +0.029$; §5.2). That sealed cell (244 queries) sits just below the program’s 250-query power floor, so its interval is appropriately wider than the development pin’s; the sealed LongDocURL documents remain unevaluated (out of scope for this MMLongBench-scoped result).

- **One cell of two.** The beat is scoped to the dispersed-evidence regime; LongDocURL is a matched hold, not a win.
- **Single reader.** All results are a Qwen3-VL-8B case study; generalization to other VLMs is untested.
- **Budget semantics.** The PAQ-KV budget is a nominal 5% decode-attention budget (realized ≈ 5.9 – 6.0%) with block K/V computed under full-document prefill context—Quest/SnapKV claim semantics, not a matched-token prefill comparison.
- **No speed advantage (measured, not deferred).** §5.5 measures the deployment profile on a systems subset: PAQ-KV is $\approx 2.25\times$ slower per query and far heavier in resident state ($\approx 730\times$) than ColQwen2, and only $\approx 11\times$ faster than full context. The efficiency argument is simplicity and deployment shape (no separately trained retriever, no index, one model, reversible query-time selection), *not* throughput; the mask route is the accuracy path and physically gathering the cache for memory savings is future work.

7 Conclusion

PAQ-KV is an architecture that turns a frozen VLM’s own KV cache into a queryable, reversible evidence store: pre-fill once, then per query re-select below-page blocks and decode under a restricted-attention 4D mask over the intact cache, with no gather and M-RoPE preserved. Reversibility is the property this design unlocks, and it holds across scorers and granularities. On dispersed-evidence long-document VQA the full method beats a state-of-the-art trained visual retriever at matched budget and is statistically indistinguishable from the gold-page oracle at $\sim 6\%$ of context, with the mechanism adversarially audited. Where it does not beat the retriever (LongDocURL) we report the matched result as a limitation and characterize the cell as apparently conversion-bound. The beat is **confirmed out-of-sample** on a fresh, disjoint, sealed held-out set evaluated exactly once ($+0.076$, $LB_{95} = +0.029$), and the dispersed-vs-conversion-bound split is offered as a hypothesis about where below-page reversible selection helps, to be tested on further datasets.

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Group	Arm	LDU	MMLB
ours (page level)	PAQ (fresh)	0.386	0.258
	attn_max (alt-agg)	0.337	0.152
isolation	same-scorer commit-once	0.128	0.095
	gold-informed static	0.278	0.154
retrieval	ColQwen2-fresh	0.512	0.328
	ColQwen2-stale	0.216	0.138
	CLIP-fresh	0.155	0.164
reference	SnapKV-fresh	0.298	0.157
	full-context	0.546	0.397
	oracle-evidence	0.609	0.431
query-blind	SnapKV-stale	0.083	0.074
	H2O	0.045	0.039
	FastV (proxy)	0.045	0.031
	LOOK-M ($\rightarrow$ H2O)	0.045	0.039
	KVzip (proxy)	0.047	0.048
	random	0.068	0.022
	truncation	0.033	0.041

Table 3: Every executed arm, accuracy at $b=5\%$, per cell (LDU = LongDocURL, MMLB = MMLongBench-Doc), on the **903-query confirmatory panel** (paq_verdict). Query-aware arms marked fresh re-select per query; stale arms commit once per document. Cohort note: this 903-query confirmatory panel is the page-level program with the *uncorrected* ColQwen2 loader (ColQwen2-fresh MMLB 0.328); the below-page headline (Table 1) is a distinct top-up cohort (444 decoded / 440 paired) scored with the *corrected* loader (ColQwen2-fresh MMLB 0.3282, §4.4). The loader correction is immaterial, so the shared reference rows agree to rounding.

A Full Executed Panel

Table 3 reports every executed arm at the gate bud-
get on the confirmatory pin. The query-blind eviction
family sits at or below random gold-page recall on
these cells—the anti-strawman record: beating it is vis-
ibly not the claim. Secondary pooled statistics: the fresh
page-level selector beats the best family arm (SnapKV-
stale) with min-over-family $LB_{95} = +0.202$; fresh –
SnapKV-fresh = $+0.094 [+0.069, +0.122]$; fresh – or-
acle = $-0.199 [-0.239, -0.161]$. Gold-page recall at
 $b=5\%$ (LongDocURL / MMLongBench-Doc): fresh page
selector 0.494/0.443; gold-informed static 0.381/0.373;
same-scorer commit-once 0.137/0.148; ColQwen2-fresh
0.664/0.536.

B The Page-Level History in Detail

All numbers in this section are on the **903-query confir-
matory cohort** (page-level arms), where ColQwen2-fresh
scores 0.512/0.328 (LDU/MMLB) with the uncorrected
loader; this is a distinct cohort from the below-page top-up
used in the headline (where the corrected-loader ColQwen2-
fresh MMLB is 0.3282, §4.4). The loader correction is im-
material (§4.4), so these page-level parity conclusions are un-
affected.

Deployable retrieval comparison @ $b=5\%$ — PAQ loses to ColQwen2-fresh; reversibility (fresh > stale) holds for BOTH scorers

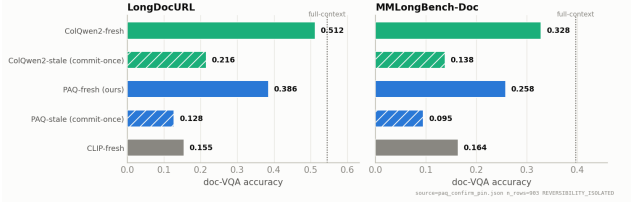


Figure 2: The two honest facts of the page-level history, per cell: (i) ColQwen2-fresh beats the page-level attention router (the ≈ 10 -point loss); (ii) *both* scorers collapse when committed once—reversibility is a general property of the multi-query regime, not scorer-specific.

The page router (PAQ v1). Pages are rendered at coarse resolution ($\approx 3.5K$ visual tokens total, i.e. ≈ 70 tokens/page on a 50-page document) and prefilled once; per query, the question rows’ attention (mean over heads, layers, and question tokens) scores pages; the top pages at budget b are re-rendered at full resolution and read. This is page routing, not KV compression, and it claims no memory savings.

The retrieval loss. At $b = 5\%$, ColQwen2-fresh scores 0.512/0.328 vs. PAQ 0.386/0.258: pooled -0.099 $[-0.129, -0.072]$ —an ≈ 10 -point loss that holds on every stratum examined (Figure 2). Gold-page recall is consistent with a coverage contribution: ColQwen2-fresh 0.664/0.536 vs. PAQ 0.494/0.443. Both scorers collapse when committed once (ColQwen2 stale 0.216/0.138), reproducing the reversibility pattern with a second scorer family.

PAQ-T: expert-head recall and the parity read-eval. Scoring with a single expert (layer, head) chosen by held-out-document cross-validation—CAPS’s technique—lifts held-out gold-page recall@5% to 0.683 ($n=451$)/0.626 ($n=425$), meeting or exceeding ColQwen2’s in-harness 0.664/0.536; mean aggregation at higher scoring resolution moves almost nothing (0.494/0.443 $\rightarrow$ 0.500/0.452), so the head, not resolution, is the lever. All selected fold heads sit in layer 21. The pre-registered accuracy read-eval (bars frozen before the run; same reader, render policy, and scorer as the cached baseline rows; only the selected page set differs) lands at **parity**: PAQ-T 0.518/0.357 vs. ColQwen2-fresh 0.512/0.328; deltas $+0.006$ $[-0.018, +0.030]$ and $+0.028$ $[+0.001, +0.055]$; pooled $+0.017$ $[-0.000, +0.035]$ —non-inferior under the frozen $LB_{95} > -0.03$ margin on both cells, but not the pre-registered beat (all $LB_{95} > 0$), and we do not present the razor’s-edge per-cell interval as a win. The initially better-looking complete-case verdict (pooled $+0.023$, $LB_{95} = +0.005$) was *blocked* in review for informative missingness—the $\approx 4\%$ of queries whose high-resolution scoring ran out of memory had cached baseline accuracy 0.62, far above the cell means—and a conservative reduced-resolution recovery of 31/35 missing rows (degrading only the PAQ-T arm, never the baseline) erased the pooled edge. Two caveats carry over: head selection is cross-validated *benchmark-supervised* model selection (not zero-

Reversibility — the axis this program tests

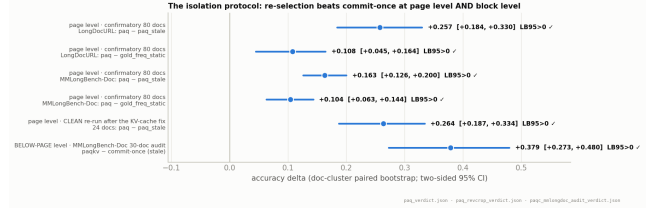


Figure 3: Reversibility—the tested axis. **Top:** a commit-once deployment reuses one subset for every question and goes stale; re-selection re-derives the subset per query from the same retained prefill. **Bottom:** the isolation protocol across granularities: the confirmatory page-level contrasts, the clean re-run after the cache fix, and the block-level PAQ-KV contrast. Every interval clears zero.

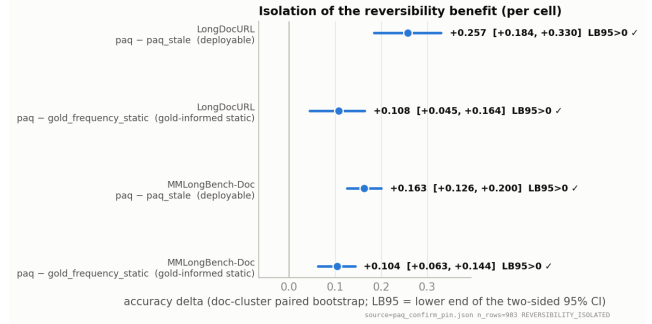


Figure 4: Per-cell forest of the two page-level isolation contrasts at $b = 5\%$. Intervals are document-clustered two-sided 95% CIs; all four exclude zero.

shot; the externally selected L21.H18 of the STOP ablation later removed this caveat from the head’s existence, not from its benchmark-tuned use), and the expert-head technique is CAPS’s—the residual novelty at page level is only the amortized one-prefill-many-queries architecture.

C Additional Reversibility Detail

Figures 3 and 4 show the reversibility axis schematically and the per-cell forest of the confirmatory isolation contrasts. Additional pre-registered analyses: (i) *answerable subset* (secondary; conditions on a model outcome): on queries where oracle-evidence scores exactly 1 ($n = 407$), fresh 0.583 vs. commit-once 0.182 ($+0.395$ $[+0.324, +0.460]$) and vs. the static 0.382 ($+0.204$ $[+0.135, +0.266]$); (ii) *missingness*: 4 of 907 vision queries are missing due to a per-document timeout; worst-case imputation in either direction leaves the gate verdict unchanged; (iii) *budget curves*: the fresh selector dominates both static comparators at every budget in $\{2.5, 5, 10\}\%$ on both cells; at $b = 10\%$ on LongDocURL it reaches 0.518 vs. full-context 0.546—reading 10% of pages

recovers $\approx 95\%$ of full-context accuracy on that cell (not on MMLongBench-Doc: 0.317 vs. 0.397).

Cache-pollution quantification. The in-place cache bug of the main text was quantified before being fixed: on a 6-documents-per-cell blast-radius diagnostic (87 query-rows), the bug inflates gold-page recall by $\approx +0.05$ for a fixed scoring head and shifts selections (mean top- B Jaccard clean vs. polluted 0.82, drifting from 1.00 at q_0 to 0.61 at q_6), as expected for cumulative contamination. The recall gate behind the PAQ-T numbers was regenerated clean before any accuracy read (clean held-out expert recall 0.683/0.626 vs. polluted 0.674/0.628), and the clean cross-validation picks a more robust head. Absolute fresh-arm accuracy in the prefix confirmatory carries ≈ 0.04 inflation (clean – polluted -0.041 [$-0.077, -0.003$] on the 24-document check); the isolation contrasts cancel the bug and stand.

D Reproducibility and Protocol Notes

Pre-registration discipline. Every gate in this program was frozen (bars, comparators, budgets, imputation policy) before its run: the two-comparator isolation gate; the PAQ-T read-eval bars (non-inferiority/beat/inferior margins, per-cell power floors, adversarial missingness imputation); the per-head-mixture transfer gate; the PAQ-KV decode-identity kill criterion, screen gates, per-cell beat bars, and the sharp-scorer (V1) gate. Two early positives did not survive adversarial review and were retracted before this draft: a first isolation gate against the eviction family (invalidated by a weak-static artifact and replaced by the two-comparator gate), and the “beats the gold-page oracle” phrasing (corrected to “matches” at power). The commitment-pressure mechanism hypothesis failed its formal test and is reported as exploratory.

Instrumentation. All significance statements use one instrument: the document-cluster paired bootstrap (10,000 draws, seed 0, percentile intervals; equal-cell pooling), unit-tested alongside the gate and pairing logic. Evaluation pins are sha-256 checksummed and byte-verified before use; the screen pin is a strict prefix of the confirmatory pin, and independent-extension analyses use the 50 new documents per cell only. Figures are generated from the verdict artifacts by a single script; nothing is hand-entered. Per-query, per-arm rows, frozen-bar metadata, verdict files, and the adversarial review trail are retained and will be released with the code.

Sealed one-touch protocol. The 80-document confirmatory pin has been reused across this program and is therefore development-only. The fresh sealed set (zero overlap with the pin; 45 MMLongBench-Doc documents / 244 queries, plus 73 LongDocURL documents left unevaluated as out of scope) was evaluated *exactly once* under a guarded, frozen harness: the confirmatory bar (complete-case document-clustered $LB_{95} > 0$, with an adversarial worst-case-imputation fallback binding only under any missingness), the corrected-loader comparator, the 5% budget, the seed, and sha-256 hashes of all eight code components plus the sealed pin were frozen before the run;

Algorithm 1 PAQ-KV: per-query reversible below-page KV-block selection with a 4D restricted-attention mask

Require: document pages rendered as in the full-context arm ($\leq 64K$ tokens); queries $q_1, \dots, q_k$; nominal budget $b = 5\%$; frozen VLM

- 1: prefill the document once; retain the full KV cache C (context length n_{ctx}); record page-aligned 64-token block spans $B_1, \dots, B_m$ and the always-keep scaffold set A
- 2: **for** each query q_j **do**
- 3: forward the tokenized question over C ; capture question $\rightarrow$ context attention via per-layer hooks; **crop** C back to n_{ctx}
- 4: $\text{score}(B_i) \leftarrow$ aggregated question-row attention mass on block B_i ’s positions
- 5: $S_j \leftarrow$ top blocks by score at the nominal budget $b \cdot n_{\text{ctx}}$
- 6: build the 4D additive mask M_j : decode rows may attend to $A \cup S_j \cup q_j$ generated tokens; all other context positions get $-\infty$
- 7: decode the answer from C under M_j —no gathering: K/V positions and M-RoPE indices unchanged
- 8: discard M_j ; C is intact for q_{j+1}
- 9: **end for**

	LongDocURL	MMLongBench
Documents	80	80
Complete queries at gate	463	440
Gold-bearing queries / doc	5.79	5.50
Median (mean) union pages	44 (45.0)	28 (36.5)

Table 4: The confirmatory evaluation pin. “Complete queries” are rows where every gated arm scored; 4 of 907 vision queries are missing due to a per-document timeout, and worst-case imputation leaves every gate verdict unchanged.

the harness refuses the sealed pin absent an explicit authorization flag and a typed confirmation, aborts on any component-hash mismatch, and runs a blind non-scoring preflight that can abort without consuming the test. The single run returned the program’s one mandate-grade claim: PAQ-KV@5% 0.396 vs. ColQwen2-fresh 0.320, $+0.076$, document-clustered $LB_{95} = +0.029$ (244/244 complete, zero skips, adversarial-robust; §5.2). The verdict, per-query rows, and a provenance manifest (pin/component hashes, resolved model revisions) are retained and released with the code.

E Moved Method and Gate Tables

F Screen-Only Budget Sweep

The beat is not a 5%-only artifact: PAQ-KV at 2.5%, 5%, and 10% all exceed ColQwen2@5% ($+0.103, +0.077, +0.104$; the 2.5/10% points are on the 157-query screen subset and the 5% point on the full 440), so although PAQ-KV’s own budget curve is mildly non-monotone here, every budget clears the retriever. A full-pin ColQwen2 re-read at 2.5/10%

Contrast at $b=5\%$	LongDocURL	MMLongBench-Doc	pooled (equal-cell)
fresh — same-scorer commit-once	+0.257 [+0.184, +0.330]	+0.163 [+0.126, +0.200]	+0.210 [+0.168, +0.250]
fresh — gold-informed static	+0.108 [+0.045, +0.164]	+0.104 [+0.063, +0.144]	+0.106 [+0.067, +0.141]
Verdict	REVERSIBILITY ISOLATED (all $LB_{95} > 0$, both cells and pooled)		

Table 5: Confirmatory isolation at $b=5\%$ on 903 complete queries (accuracy; document-cluster paired bootstrap; rows report $\Delta [LB_{95}, UB_{95}]$). The two-comparator gate was frozen before the run.

is future work.

G Additional Limitations Detail

LongDocURL conversion-bound decomposition. A re-analysis of the executed rows *suggests* the shortfall there is largely past selection: of the 463 rows, $\approx 39\%$ fail under both PAQ-KV and the gold-page oracle (zero even when handed the native-resolution gold pages), only $\approx 10\%$ are oracle-right/PAQ-KV-wrong (the selection-capturable pool), and 2.2% are PAQ-KV-right/oracle-wrong—so the oracle is not a strict ceiling and “no selection method can rescue these” is an observation on this dataset, not a law. Native resolution already saturates the prefill on 79/80 documents, and large documents are a flat, low-leverage half. Among pure-selection levers only a knife-edge oracle-chooser union clears ($LB_{95} = +0.003$, undeployable); only a $PAQ-KV \cup ColQwen2$ hybrid clears comfortably ($LB_{95} = +0.061$) and we exclude it as not a standalone method. We therefore offer the dispersed-vs-conversion-bound split as a *hypothesis* from two datasets.

Base-model ceiling counts. A large fraction of queries receive no credit even given exactly the annotated gold pages: the oracle-evidence arm scores 0 on $317/903 = 35.1\%$ of confirmatory queries and a full 1 on only 45.1% . Pairing controls for shared query difficulty; the oracle is *not* a strict ceiling (a selected page set can beat the annotated gold set on individual queries). The query-blind eviction arms are disclosed proxies we claim nothing against.

Protocol residues. The commit-once aggregate uses the first $M = \min(3, k)$ queries in frozen arrival order (order-permutation and M -sensitivity checks remain to be run; the order-free static partially covers this), and the page-level confirmatory run predates the cache fix (§4.4)—its isolation *contrasts* were verified robust, but its absolute fresh-arm levels carry ≈ 0.04 inflation on the 24-document check.